

Negative Moments of Hilbert-Kernel Weights

A full candidate solution to the moment conjecture in arXiv:2106.03354

Automated mathematics run `fa_banach_001`, lane 2

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Status. Full solution candidate, likely valid, awaiting human review. The theorem proves the asymptotic formula and the necessary-and-sufficient integrability condition explicitly conjectured for every moment order $-1 < \beta < 0$. It does not claim existence of the separate heuristic scaling density p .

1 Source problem

The source is P. P. Mitra and C. Sire, *AI without networks*, arXiv:2106.03354 [1]. Let $X_0, \dots, X_n$ be independent random vectors in $\mathbb{R}^d$ with probability density ρ . At a fixed point x with $\rho(x) > 0$, the Hilbert-kernel weights are

$$w_i(x) = \frac{\|x - X_i\|^{-d}}{\sum_{j=0}^n \|x - X_j\|^{-d}}. \quad (1)$$

For a real parameter γ , put

$$\kappa_\gamma(x) = \int_{\mathbb{R}^d} \frac{\rho(x+y)}{\|y\|^{\gamma d}} dy, \quad (2)$$

whenever this extended integral is finite. If $\gamma < 0$, this means

$$\kappa_\gamma(x) = \int_{\mathbb{R}^d} \rho(x+y) \|y\|^{|\gamma|d} dy.$$

Write V_d for the volume of the Euclidean unit ball in $\mathbb{R}^d$.

Theorem 1 of the source gives, for $0 < \beta < 1$,

$$\mathbb{E}[w_0(x)^\beta] \sim \frac{\kappa_\beta(x)}{(V_d \rho(x) n \log n)^\beta},$$

and gives only an upper bound when $-1 < \beta < 0$. Page 31 conjectures that the same equivalent remains valid for the negative moments whenever they exist, and that $\kappa_\beta(x) < \infty$ is also necessary.

The source context defining the target formula and sufficient condition is reproduced below.

- The expected value of the entropy $S(x)$ (moment for “ $\beta = 1^-$ ”) behaves as,

$$\mathbb{E}[S(x)] = (n+1) \mathbb{E}[-w_0(x) \ln(w_0(x))] \underset{n \rightarrow +\infty}{\sim} \frac{1}{2} \ln(n). \quad (28)$$

- For $0 < \beta < 1$: defining $\kappa_\beta(x) := \int \frac{\rho(x+y)}{\|y\|^\beta} d^d y < \infty$, we have

$$\mathbb{E}[w_0^\beta(x)] \underset{n \rightarrow +\infty}{\sim} \frac{\kappa_\beta(x)}{(V_d \rho(x) n \ln(n))^\beta}. \quad (29)$$

- For $\beta < 0$: all moments for $\beta \leq -1$ are infinite, and the moments of order $-1 < \beta < 0$ satisfy

$$\mathbb{E}[w_0^\beta(x)] \leq 1 + n \kappa_{|\beta|}(x) \kappa_\beta(x), \quad (30)$$

so that a sufficient condition for its existence is $\kappa_\beta(x) = \int \rho(x+y) \|y\|^{|\beta|d} d^d y < \infty$.

Figure 1: Theorem 1 and equations (29)–(30) in the source, PDF page 29.

$$\underset{n \rightarrow +\infty}{\sim} \frac{\int_0^{+\infty} p(w) w^\beta dw}{(n \ln(n))^\beta}, \quad (43)$$

where the last integral converges since $p(w) \underset{w \rightarrow +\infty}{\sim} w^{-2}$ and $\beta < 1$. This result is perfectly consistent with Eq. (29) in Theorem 1, and suggests that $\int_0^{+\infty} p(w) w^\beta dw = \frac{\kappa_\beta(x)}{(V_d \rho(x))^\beta}$. Interestingly, for $0 < \beta < 1$, and contrary to the case $\beta > 1$, we find that the large n equivalent of the moment is not universal and depends explicitly on x and the density ρ . As for moments of order $-1 < \beta < 0$, we conjecture that they are still given by Eq. (29) (and equivalently, by Eq. (43)) provided they exist, and that the sufficient condition for their existence $\kappa_\beta(x) < \infty$ is hence also necessary, since $\kappa_\beta(x)$ also appears in Eq. (29). The fact that moments for $\beta \leq -1$ do not exist strongly suggests that $p(0) > 0$. In fact, Eq. (43) also suggests that all moments for $-1 < \beta < 0$ exist if and only if $0 < p(0) < \infty$. In Fig. 1b (see also the last section of the appendix A.2), we present numerical simulations confirming our scaling ansatz, the fact that $p(w) \underset{w \rightarrow +\infty}{\sim} w^{-2}$, and the quantitative prediction for W_n .

Figure 2: The explicit negative-moment conjecture in the source, PDF page 31.

2 Proof intuition

Set $\beta = -a$ with $0 < a < 1$ and

$$Y_i = \|x - X_i\|^{-d}, \quad S_n = Y_1 + \cdots + Y_n.$$

Continuity and positivity of ρ at x imply the borderline heavy-tail law

$$\mathbb{P}(Y_i > t) \sim \frac{V_d \rho(x)}{t}.$$

Thus S_n has the deterministic first-order scale $V_d \rho(x) n \log n$. Although the summands have infinite mean, truncation at this scale gives a weak law. A tail-integral estimate supplies uniform integrability for every power below one, upgrading the weak law to

$$\mathbb{E}[S_n^a] \sim (V_d \rho(x) n \log n)^a.$$

The weight ratio then separates the only two random pieces:

$$w_0(x)^{-a} = (1 + \|x - X_0\|^d S_n)^a.$$

Because $0 < a < 1$,

$$z^a \leq (1 + z)^a \leq 1 + z^a.$$

Independence sandwiches the desired expectation between $\kappa_{-a}(x) \mathbb{E}[S_n^a]$ and one plus that quantity. The same identity also makes existence exact: $\kappa_{-a} < \infty$ is sufficient by subadditivity, while one term of S_n proves necessity.

3 Main theorem

Theorem 1 (Negative-moment conjecture). *Let $X_0, \dots, X_n$ be i.i.d. random vectors in $\mathbb{R}^d$ with probability density ρ , and let $x \in \mathbb{R}^d$ satisfy $\rho(x) > 0$ and be a point of continuity of ρ . Define the Hilbert weights by (1). For every $-1 < \beta < 0$ and every $n \geq 1$,*

$$\mathbb{E}[w_0(x)^\beta] < \infty \iff \kappa_\beta(x) < \infty. \quad (3)$$

When these equivalent conditions hold,

$$\mathbb{E}[w_0(x)^\beta] \underset{n \rightarrow \infty}{\sim} \frac{\kappa_\beta(x)}{(V_d \rho(x) n \log n)^\beta}. \quad (4)$$

The proof uses the following elementary borderline heavy-tail lemma.

Lemma 2 (Fractional moments at tail index one). *Let $Y, Y_1, Y_2, \dots$ be i.i.d. nonnegative random variables and suppose*

$$\mathbb{P}(Y > t) \sim \frac{c}{t} \quad (t \rightarrow \infty) \quad (5)$$

for some $c > 0$. If $S_n = Y_1 + \cdots + Y_n$ and $b_n = cn \log n$, then

$$\frac{S_n}{b_n} \longrightarrow 1 \quad \text{in probability.}$$

Moreover, for every $0 < a < 1$,

$$\mathbb{E}[S_n^a] \sim b_n^a. \quad (6)$$

Proof. Let $\bar{F}(t) = \mathbb{P}(Y > t)$. For $b > 0$, the layer-cake identity gives

$$m(b) := \mathbb{E}[Y \mathbf{1}_{\{Y \leq b\}}] = \int_0^b \bar{F}(t) dt - b\bar{F}(b) \sim c \log b. \quad (7)$$

Also, using $\bar{F}(t) \leq C/t$ for all sufficiently large t ,

$$\mathbb{E}[Y^2 \mathbf{1}_{\{Y \leq b\}}] \leq 2 \int_0^b t \bar{F}(t) dt = O(b). \quad (8)$$

Put $T_n = \sum_{i=1}^n Y_i \mathbf{1}_{\{Y_i \leq b_n\}}$. Since $\log b_n / \log n \rightarrow 1$, (7) yields

$$\frac{\mathbb{E}T_n}{b_n} = \frac{nm(b_n)}{cn \log n} \rightarrow 1.$$

Equation (8) gives

$$\frac{\text{Var}(T_n)}{b_n^2} \leq \frac{n O(b_n)}{b_n^2} = O\left(\frac{1}{\log n}\right) \rightarrow 0.$$

Finally,

$$\mathbb{P}(S_n \neq T_n) \leq n\bar{F}(b_n) \sim \frac{1}{\log n} \rightarrow 0.$$

Chebyshev's inequality therefore proves $S_n/b_n \rightarrow 1$ in probability.

It remains to justify convergence of fractional moments. Fix any $0 < q < 1$. For t large,

$$\mathbb{E}[Y \mathbf{1}_{\{Y \leq t\}}] \leq C(1 + \log t).$$

If $S_n > t$, either some $Y_i > t$, or the sum of the variables truncated at t exceeds t . A union bound and Markov's inequality give

$$\mathbb{P}(S_n > t) \leq n\bar{F}(t) + \frac{n}{t} \mathbb{E}[Y \mathbf{1}_{\{Y \leq t\}}] \leq \frac{Cn(1 + \log t)}{t}. \quad (9)$$

For all sufficiently large n , integrate this estimate only above b_n :

$$\begin{aligned} \mathbb{E}[S_n^q] &= q \int_0^\infty t^{q-1} \mathbb{P}(S_n > t) dt \\ &\leq b_n^q + Cqn \int_{b_n}^\infty t^{q-2} (1 + \log t) dt \\ &\leq b_n^q + C_q n b_n^{q-1} (1 + \log b_n) \leq C'_q b_n^q. \end{aligned}$$

Here the last inequality follows because $n(1 + \log b_n)/b_n$ remains bounded.

Now fix $0 < a < 1$ and choose q with $a < q < 1$. The preceding bound says that $\{(S_n/b_n)^a\}$ is bounded in $L^{q/a}$, hence is uniformly integrable. Since it converges in probability to 1, its expectations converge to 1. This is precisely (6). $\square$

Proof of Theorem 1. Write $\beta = -a$, where $0 < a < 1$, and set

$$R_i = \|x - X_i\|, \quad Y_i = R_i^{-d}, \quad S_n = \sum_{i=1}^n Y_i.$$

The variables R_0 and S_n are independent. Moreover,

$$\begin{aligned} t\mathbb{P}(Y_1 > t) &= t \int_{\|y\| < t^{-1/d}} \rho(x+y) dy \\ &\longrightarrow V_d \rho(x) \end{aligned}$$

by continuity of ρ at x . Thus Lemma 2, with $c = V_d \rho(x)$, gives

$$\mathbb{E}[S_n^a] \sim (V_d \rho(x) n \log n)^a. \quad (10)$$

We first settle existence. Local boundedness of ρ at x and $a < 1$ show that

$$\kappa_a(x) = \mathbb{E}[R_1^{-ad}] < \infty :$$

the singularity $\|y\|^{-ad}$ is locally integrable in $\mathbb{R}^d$, and away from the origin it is bounded. If $\kappa_{-a}(x) < \infty$, then

$$S_n^a \leq \sum_{i=1}^n Y_i^a \quad \text{and} \quad (1+z)^a \leq 1+z^a$$

for $z \geq 0$. Consequently,

$$\begin{aligned} \mathbb{E}[w_0(x)^{-a}] &= \mathbb{E}[(1 + R_0^d S_n)^a] \\ &\leq 1 + \mathbb{E}[R_0^{ad}] \mathbb{E}[S_n^a] \\ &\leq 1 + n \kappa_{-a}(x) \kappa_a(x) < \infty. \end{aligned}$$

Conversely, for $n \geq 1$,

$$w_0(x)^{-a} = (1 + R_0^d S_n)^a \geq (R_0^d Y_1)^a.$$

Tonelli's theorem and independence imply

$$\mathbb{E}[(R_0^d Y_1)^a] = \mathbb{E}[R_0^{ad}] \mathbb{E}[R_1^{-ad}] = \kappa_{-a}(x) \kappa_a(x).$$

Because $0 < \kappa_a(x) < \infty$, the expectation is infinite whenever $\kappa_{-a}(x) = \infty$. This proves (3).

Assume now that $\kappa_{-a}(x) < \infty$. The elementary two-sided bound

$$z^a \leq (1+z)^a \leq 1+z^a$$

and independence give

$$\kappa_{-a}(x) \mathbb{E}[S_n^a] \leq \mathbb{E}[w_0(x)^{-a}] \leq 1 + \kappa_{-a}(x) \mathbb{E}[S_n^a].$$

Divide by $(V_d \rho(x) n \log n)^a$ and use (10). We obtain

$$\mathbb{E}[w_0(x)^{-a}] \sim \kappa_{-a}(x) (V_d \rho(x) n \log n)^a.$$

Replacing $-a$ by β is exactly (4). □

4 Verification notes

- The tail constant is checked directly: $\{Y > t\}$ is the ball of radius $t^{-1/d}$ about x , whose probability is asymptotic to $\rho(x)V_d t^{-1}$.
- The only delicate passage from a weak law to an expectation is made explicit in Lemma 2: an L^q bound for some $q > a$ gives the required uniform integrability.
- Both directions of the integrability criterion hold for each fixed $n \geq 1$; they are not merely asymptotic.
- The source’s upper bound (30) is recovered verbatim from $(1 + z)^a \leq 1 + z^a$. Necessity comes from the matching lower bound using just the first summand of S_n .
- The script `code/check_uniform_example.py` performs a finite Monte Carlo check for the one-dimensional uniform density. It is only a sanity check and is not used in the proof.

5 Novelty search and limitations

The bounded novelty check was performed on 23 July 2026. The run indexes were searched for arXiv:2106.03354, “Hilbert kernel”, “negative moments”, and the $n \log n$ fractional-moment mechanism. Web/arXiv searches used the exact conjecture phrase, the notation $\kappa_\beta(x)$, the authors’ names, and close variants involving tail-index-one sums. These searches found the source and its repository copies, plus general heavy-tail papers, but no paper claiming this negative-moment asymptotic or the necessary-and-sufficient condition. The latest local arXiv source/PDF, dated 7 June 2024, still contains the conjecture. This is a bounded search, not an exhaustive specialist bibliography, so novelty remains subject to expert review.

The theorem does not prove the source’s separate heuristic claim that the rescaled weights possess a limiting density p , nor any assertion about $p(0)$. It also does not treat discontinuity or vanishing of ρ at x . The range $\beta \leq -1$ is outside the target because the source already proves those moments infinite.

6 Human-review recommendation

Recommended disposition: **review as a likely full solution** to the explicit negative-moment conjecture. The primary verifier focus should be Lemma 2, especially the tail-integral bound (9) and the uniform-integrability step. A secondary check should confirm that the source’s notation uses n other samples in the denominator of w_0 , as in (1); an $n + 1$ convention would only make an asymptotically negligible change.

References

- [1] P. P. Mitra and C. Sire, *AI without networks*, arXiv:2106.03354, latest source consulted dated 7 June 2024.